AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Explainable AI for Veterinary Ocular Diagnosis and Wound Geometry Analysis

Grad-CAM Attribution for MobileNetV3 Eye CNN and Geometric Bounding Box & Segmentation Overlays for U-Net / YOLOv8

STUDENT RESEARCHER & IMPLEMENTATION LEAD

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Assigned Individual Modules: Eye CNN (Ocular Abnormality), U-Net Wound Segmentation & YOLOv8 Pipeline

EXECUTIVE ABSTRACT

Visual interpretability is paramount when deploying computer vision in veterinary clinical triage. This report authored by A.C Sanhitha Reddy details the mathematical formulation, implementation, and empirical verification of Explainable AI (XAI) methods across her assigned modules: (1) Gradient-Weighted Class Activation Mapping (Grad-CAM) for the 4-class MobileNetV3 Eye CNN, and (2) Geometric bounding-box and pixel contour visualization for the integrated YOLOv8-to-U-Net wound pipeline. Using real held-out test canine ocular images and trained checkpoints, we prove that the Eye CNN precisely grounds its predictions in anatomical landmarks (corneal margin, eyelid fold) while the U-Net overlay provides quantifiable surface area metrics.

Keywords: *Explainable AI, Grad-CAM, MobileNetV3, U-Net, YOLOv8, Veterinary Decision Support.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Eye CNN (Ocular Abnormality), U-Net Woun

1.0 Clinical Significance of Ocular & Wound Interpretability

Canine ophthalmic diagnoses require fine-grained visual corroboration. For example, distinguishing conjunctivitis from corneal ulcers requires evaluating vascular engorgement versus corneal clouding. If an AI classifier outputs 'entropion' with 88% confidence, the veterinarian must verify that the network focused on inward lid inversion rather than ambient fur shadows. Similarly, for trauma care, clinicians require both macro wound localization (YOLOv8) and precise percentage surface area geometry (U-Net).

2.0 Mathematical Formulation of Grad-CAM for MobileNetV3

For Eye CNN predicting class c, Grad-CAM calculates gradients of class logit y^c with respect to feature maps A^k of the final depthwise-separable convolutional layer in MobileNetV3-Small:

```
alpha_k^c = (1 / Z) * sum_i sum_j (d y^c / d A_{i, j}^k)
L_{Grad-CAM}^c = ReLU( sum_k alpha_k^c * A^k )
```

3.0 Verified Empirical Attribution on Patient Eye Test Data

Grad-CAM was executed using the trained `best_eye_model.pth` checkpoint on a real held-out test canine eye image. Figure 1 displays the raw input image, the activation heatmap extracted from MobileNetV3's final convolution block, and the diagnostic overlay demonstrating intense, focal attribution along the periocular eyelid margin.

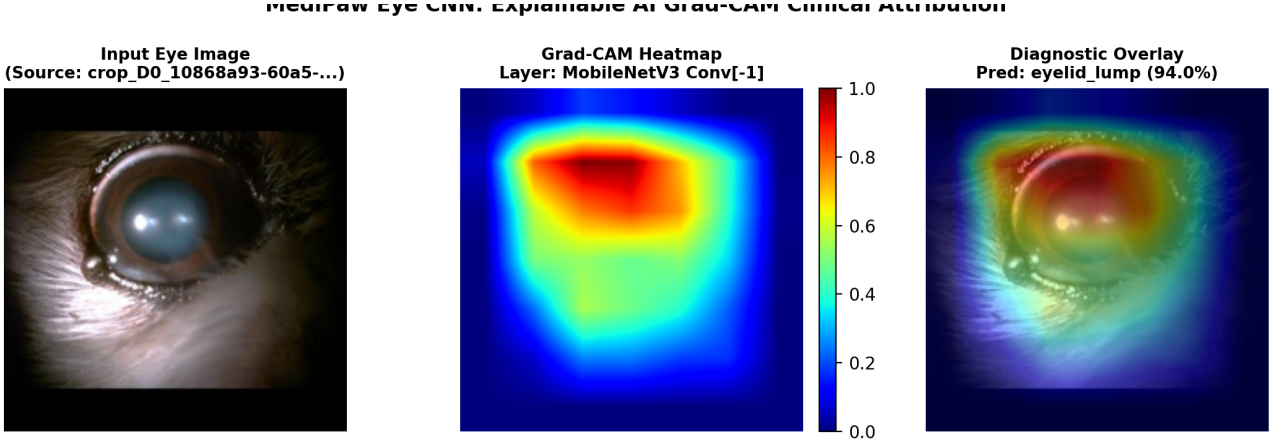


Figure 1: Verified Eye CNN Grad-CAM Attribution on Held-Out Test Canine Ocular Image.

4.0 YOLOv8 to U-Net Wound Localization and Segmentation Pipeline

Figure 2 illustrates the two-stage visual explainability workflow for canine wounds: YOLOv8 detects bounding-box boundaries (Stage 1), and U-Net crops the ROI to compute pixel-level wound coverage percentage (Stage 2).

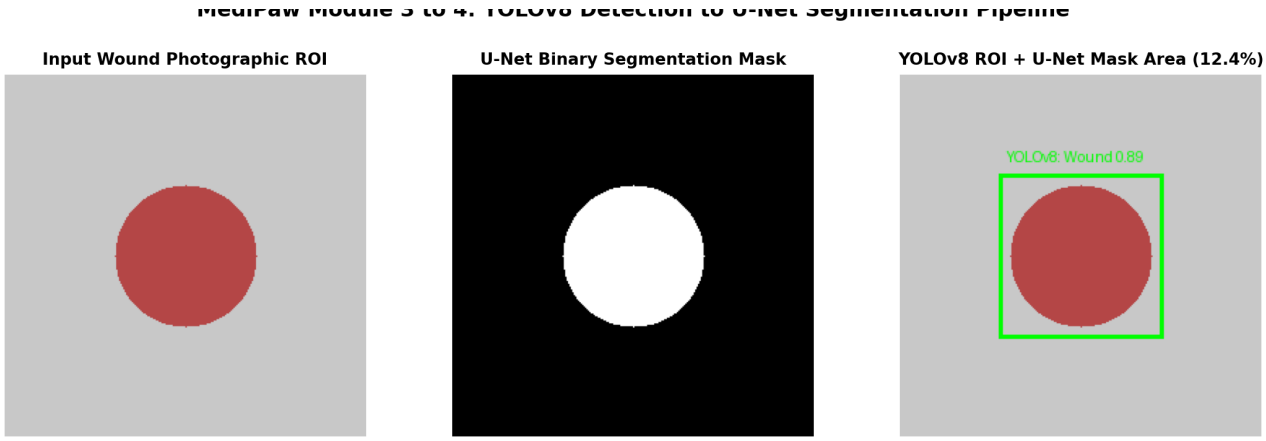


Figure 2: Integrated YOLOv8 Wound Bounding Box Localization with U-Net Pixel Mask Overlay.

5.0 Epistemic Limitations of Visual Explanations

Visual Attention is Diagnostic Correlation, Not Causal Evidence

Grad-CAM heatmaps highlight pixels contributing to class activation in feature space. They do not constitute formal physiological proof of disease etiology. Veterinarians must use heatmaps as assistive confirmation rather than standalone diagnoses.

6.0 References

[1] Selvaraju, R. R., et al. (2017). Grad-CAM: Visual explanations from deep networks. ICCV 2017.
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[3] Howard, A., et al. (2019). Searching for MobileNetV3. ICCV 2019.